Supplementary Material — AI Council: From Collaboration to Governance (v2.9.2, 2026-10-08)

Zheng Wan Independent Researcher (matt.wanzheng@gmail.com)

Tiara Wan Liberal Arts (Boya) College, Chongqing University (tiarawzy@gmail.com)

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This supplementary package accompanies the submission. Contents: Appendix A (empirical data source index), Narrow-Domain F1 Detail, Appendix C (lessons learned L7–L11), Table S1 (19 failure modes → governance module mapping), and Appendix E (control-plane integrity).

Appendix A: Empirical Data Source Index

Table 1: Appendix A: Empirical Data Source Index and Reproducibility Anchors

Paper Metric / Claim	Source (Reproducible)
Net correction −3 (session-level +9/−12; member-level +8/−11)	`data/meetings/trackm/` 24 sessions `net_correction.json` + verification report (dual-caliber comparison table)
Consensus 37.5% / dissent preservation 62.5%; D0=0/D1=100%/D2=100%	Ibid. 24 sessions `gates.json`
F1 0.06 vs 0.34 ($p<0.0001$, 87 sessions)	`data/exp/runs/e1_batch_1-3` + narrow-domain analysis report
Governance contrast four failure modes zeroed out	Run 1/Run 2 `gates.json` + `qa_report_g3_run2.md`
Coverage 33.3%→83.3%, re-query recovery rate 66.7%	`symposium_20260905_212526_b25d55/gates.json` (S1–S6 pre-registered acceptance script)
120-condition primary benchmark + 600 GT items	`data/exp/test_main_120.jsonl` + `gt/` (`gt_summary.json`, G1 gate)
Eight-model behavioral spectrum	`trackm_stepf_all_results.json` + verification report (eight-model behavioral signature archive, data file internal entries persona-01~08)
Strong-weak performance gap 31.3pp; H6 coverage containment	`trackm_stepb2/stepc2` report
361 automated test cases	`pytest backend/` (2026-09-05, 353 baseline + SPEC-84 v1.2 new 8 items)

All artifacts cited in Appendix A are maintained in the authors' internal evidence repository with immutable raw traces; the internal paths above serve as artifact identifiers within that repository. Materials are available from the authors upon request and will be released publicly upon acceptance.

E2–E6 design parameters are detailed in §5.3 declarations and Phase B experimental plan (design-ready, not fully executed).

Narrow-Domain F1 Detail

False-Positive Analysis: In the 40-commit narrow-domain evaluation (18 samples analyzed), 83% of false positives were procedural meta-discourse mis-captured by the extraction layer; filtering (zero-API replay) raised Macro F1 to 0.0698.

Appendix C: Lessons Learned L7–L11 (Preserved from Body Reductions; renumbered after body gained L6)

L7 Direct insertion of AI-generated content into frontend interfaces creates security debt. ISSUES-021: `<script>` tags appearing in agent discussions were injected verbatim into HTML reports, resulting in stored Cross-Site Scripting (XSS). *Recommendation: All AI outputs must be sanitized as untrusted inputs across every rendering pathway.*

L8 Evaluation extraction layers must distinguish "findings regarding code" from "meta-comments regarding meeting processes". In Batch 1 initial metric evaluations, 83% of the false positive (FP) pool arose because the extraction layer misclassified internal meeting discipline discussions (e.g., "C-004 rebuttal lacks direct quotation, violating R4.2") as code defect items—the model did not hallucinate; rather, the extractor failed to differentiate discourse types. Remediation via a procedural meta-discourse classifier (pure rules: claim citations, discipline rule numbers, procedural keyword lexicons) restored metric credibility upon zero-API full replay. *Recommendation: Governance mechanisms inherently generate procedural meta-discourse as an authentic facet of system behavior—evaluation extraction layers must decouple both discourse streams, archiving filtered meta-discourse separately as governance audit assets (as practiced in this study, with 488 procedural items archived).*

L9 Procedural credibility in multi-AI collaboration requires institutional enforcement. Throughout project development, multiple instances emerged where AI collaborators reported outcomes at variance with disk artifacts (mental arithmetic errors in test counts, fabricated audit verdicts, phantom citations, reversed concession roles), each falsified via independent replication and disk audits. This led to firm rules: self-reported numbers must attach raw terminal outputs; narrative conclusions must cite claim IDs; automated phantom citation detection scripts; and automated four-tier reconciliation gates (transforming "claimed counted" into "counted by script")—eliminating self-report discrepancies post-enforcement. *Recommendation: Wherever AI-generated reports enter the decision chain, treating self-reports as unverified must serve as the default discipline—the rigor with which we govern experimental data embodies the governance philosophy championed in this paper.*

L10 Full-coverage cross-examination exhibits capacity boundaries requiring morphological re-query designs. Under extensive argumentative text (56 claims), cross-examination coverage dropped to 33%: structural capacity gaps (one utterance per round targeting one claim) compounded by examination skew (over-allocation to contentious issues, zero coverage for opening claims) and context assembly flaws; short-claim meetings (24 choice-track stances) achieved 100% coverage. Targeted re-query rounds (up to one directed query round without exemption pathways) verified via smoke testing elevated coverage to 83% (66.7% recovery rate), with residual items escalated to human review—**better to escalate to humans than compromise standards**. *Recommendation: Coverage gates must scale budgets to claim morphology, with built-in targeted re-query and loud failure mechanisms (prohibiting silent execution under empty contexts).*

L11 Fallback fault-tolerance preserves pipeline execution, not deliberation quality—council completeness is a first-order variable of deliberation output. During headless pipeline acceptance, 2 of 4 council seats were disqualified due to credential expiration; the truncated council (2/4) completed 3 full sessions supported by degradation protocols (zero pipeline interruptions, all artifacts generated), yet all three sessions yielded exactly 0 consensus items;

restoring credentials to a full council (4/4) produced 3 consensus items in a single session with execution duration halved (7 minutes vs. 12–16 minutes). Case-level evidence ($n = 3$ vs. $n = 1$, confounding scale and diversity) precludes quantitative claims, yet indicates a clear vector: **for deliberation systems, pipeline availability is a necessary but insufficient condition—every absent member thins the evidentiary substrate until consensus loses raw material.** *Recommendation: Experimental pre-flight checks for multi-agent deliberation systems must incorporate council health gates (credential validity and degradation status checks), treating council completeness as a formalized ablation dimension rather than an operational anomaly (§5.7.4).*

Table S1: 19 Documented AI-Conclusion Failure Modes → Governance Module Combination Mapping

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Appendix E: Control-Plane Integrity — Incident Ledger, Production Case, and Protocol Evolution (v2.9.2, 2026-10-08)

Added per the Evidence-Claim Alignment Pass (2026-09-15). Case-level material; nothing in this appendix alters the frozen raw traces, ledgers, or statistics.

E.1 Incident Ledger (Meta-Governance Candidates)

Two production control-plane incidents (2026-09) are recorded as **meta-governance / infrastructure candidates** — deliberately not numbered into the 19-mode taxonomy until recurrence review:

Lessons (condensed into body §7 L6): *readiness is behavioral, not declarative; control actions must be state-validated.*

E.2 Production Case: Governed Claim-Auditing Session (Case-Level Observation)

A 45-minute governed review session over a full paper draft (2026-09-13, production pipeline; meeting ledger `eb29b6f6-8032-4510-bc7f-06b90f956cf8`, terminal state `closed`): four heterogeneous committee models (DeepSeek moderator + Gemini-3.8 + LongCat + Qwen3.8), 30 utterances, cross-examination coverage enforced, 3 soft-consensus items with dissent archived, 5 actionable tasks emitted, internal quality score 69.2/100 (engineering 19.7, domain 18.5, token 15.0, human-alignment 16.0; source: internal evaluation record of the session). Notable case-level observation: the committee independently flagged a statistical-disclosure gap in the draft (a point estimate reported without confidence interval), which was closed in revision. This case is cited as production evidence of claim-auditing behavior **at case level only** — it is not a controlled evaluation and contributes no aggregate statistics.

E.3 Protocol Evolution Direction (Design Refinement, Not a Verified Contribution)

The review-session protocol is being refined from "free statement → two review rounds → consensus" toward five stages: **Discovery → Challenge → Repair → Verification → Tasking** — the second round shifts from repeated review to a claim-repair test (flagged claim

Table 2: Table S1: 19 Documented AI-Conclusion Failure Modes to Governance Module Combination Mapping

#	Failure Mode	Governance Module Combination	Key Mechanism / Threshold
F1	Anchoring effect	Blind review isolation + moderator last + Full Reveal	Independent calls mutually invisible (D6)
F2	Conformity and sycophancy	Blind phase + R5.1 + Steelman	Conformity patterns → BLOCK without vote
F3	Hallucination and fabrication	HG-1~6 + R3.2 zero tolerance	Sourceless precise data → BLOCK
F4	Fictitious citations	CV-1~4 + EvidenceStore	Official citations must resolve to evidence records
F5	Scope creep	TopicGate + R2.2 + contract locking	Relevance <0.20 intercepted; taboo terms BLOCKED
F6	Authority amplification	FIX-A claim provenance	0 citations flagged "[Moderator New Claim]"
F7	Belief contagion	FIX-B fact gate + evidence grading	Unverified assertion warnings pre-convergence
F8	Equal-weight dilution	FIX-C authority weighting	Citation/evidence rates in weights; red lines
F9	Minority dissent suppression	FIX-D + single-agent veto	High-evidence vetoes → human_review
F10	Defensive antagonism	FIX-E + Steelman	"Objective input, not criticism" injected
F11	Unsubstantiated high confidence	CV-4 + engineering_veto	conf ≥0.9 without evidence → score zeroed
F12	Knowledge-cutoff blind spots	Phase 0 declaration + R3.3	Mandatory cutoff declaration; [UNCERTAIN] tag
F13	Logical silos	R5.2 + business_logic_veto	Cross-examination must target first-round claims
F14	Budget overrun	TokenGuardian + budget_veto	Four-tier breaker; >90% daily limit → veto
F15	Lack of traceability	ClaimLedger + TraceLogger + provenance	Audit trail incl. intercepted utterances
F16	Non-compliant utterances	ComplianceEngine + dual review	BLOCK rollback ≤3; 3 strikes → deactivation
F17	Lack of accountability	HumanGate + 5 checkpoints + ≥85	D8 never auto-submits; 9 human signals
F18	Forced convergence	DegradationProtocol + alignment_veto + HumanGate	Six-source scan; conf <0.50 → open_issues
F19	Expert mismatch / roster noise	TaskClassifier + NodeSampler + ScoreMatrix + MessagePassing	Dynamic top-k admission; trust pruning; directed passing

Table 3: Table S2: Control-Plane Incident Ledger (Candidate Failure Modes)

ID	Incident	Root pattern	Mechanism response (SPEC-82/83, 244 tests green)
CP-1	Provider advertised a model as available (`/models` listing) while generation calls failed on quota/balance state	**Provider false readiness** — declared availability ≠ behavioral readiness	Preflight generation probe + quota/balance health check before meeting start
CP-2	An automated accept action fired after adjournment and restarted a closed meeting	**State-transition misuse / external automation interference** — control actions not validated against meeting phase	Expected-state transition validation, idempotency keys, phase locks, and a control-action audit log

→ owner repair → independent verification). This direction is recorded as a design refinement and future protocol extension; it has **not** been validated by a controlled experiment and supports no performance claim.